Proceedings of Machine Learning Research 329:1–6, 2026 Conformal and Probabilistic Prediction with Applications

Zero-Shot Probabilistic Stock Returns Forecasting with Pretrained RVFL Networks

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Editor: Ernst Ahlberg, Ulf Johansson, Henrik Boström, Alberto Carlevaro, Johan Hallberg Szabadváry and Lars Carlsson

Abstract

We ask whether a **zero-shot pretrained quasi-randomized functional link network** (Moudiki et al., 2018) (QRVFL) for univariate time series can match a student-t eGARCH model at multi-step probabilistic stock return forecasting, without explicitly specifying volatility dynamics. Using formal equivalence tests across 15 equity time series of length equal to 500 days, three forecast horizons ($h \in \{5, 10, 21\}$ trading days ahead), and a rolling window cross validation (with 400 training days), we show that the RVFL is statistically equivalent to eGARCH on point accuracy and distributional calibration (within given margins), while producing *slightly tighter* 95 % prediction intervals and running 10× faster than the eGARCH implementation (with simulations enabled for both models) from R package **rugarch**.

Keywords: conformal prediction, probabilistic forecasting, stock returns, eGARCH, pretrained models, equivalence testing, CRPS

1. Setup

Four models are evaluated [1]: an eGARCH model with Student-t innovations serving as a baseline (denoted as **eGARCH-t** in results' tables 1 – 4); a quasi-randomized functional link network for time series (QRVFL, (Moudiki et al., 2018)) **pretrained** (described in section 2 and implemented in (Moudiki, 2026)) on synthetic GARCH(1,1) returns to minimize out-of-sample continuous ranked probability score (CRPS) (Jordan et al., 2019) (denoted as **pretrained-rvfl2** in results' tables); and sequential split-conformal wrappers (Vovk et al., 2005; Moudiki, 2025b) around each (respectively denoted as **Conf-eGARCH-t**, **Conf-pretrained-rvfl2** in results' tables). The conformal procedure, adapted from (Moudiki, 2025b) and implemented in (Moudiki, 2026), uses a stationary block bootstrap on scaled calibration residuals to form the predictive ensemble. 15 equity series are used, yielding $n = 45$ ticker–horizon observations per model after time-series cross-validation. Evaluation covers three pillars: point accuracy (RMSE); probabilistic calibration (Coverage, CRPS, Winkler95 score); and computation time. The eGARCH baseline is competitive. (Hansen and Lunde, 2005) showed, indeed, that few volatility models reliably outperform a GARCH(1,1), **making equivalence with eGARCH-t a non-trivial result** – and one that is sensitive to the choice of margin, as we show in the appendix. Achieving statistical equivalence to eGARCH-t on both point accuracy and calibration is meaningful, precisely because the QRVFL requires no explicit volatility model specification, no iterative likelihood optimization, and runs an order of magnitude faster — making point accuracy and distributional calibration equivalence a favourable outcome, not a null result.

1. Code available at: `https://github.com/thierrymoudiki/2026_05_28_Pretrain_Ridge2_Stocks_Full_Pipeline/tree/main`

2. Pretraining

The QRVFL's hyperparameters – number of hidden nodes, number of lags, and two regularisation parameters (λ_1, λ_2) – are selected *before* any real data is seen, by minimizing CRPS on a large synthetic benchmark. Specifically, 1000 GARCH(1,1) paths of length 500 are simulated under randomized, stationary parameters ($\sigma^2_{t+1} = \omega + \alpha * \epsilon^2_t + \beta * \sigma^2_t$, with $\omega \in [10^{-7}, 10^{-5}]$, $\alpha \in [0.02, 0.15]$, $\beta \in [0.75, 0.97]$, $\alpha + \beta < 1$), with innovations ϵ drawn from either a standard normal or a standardised Student-t distribution. For each synthetic series the model is fitted on the first 400 observations and evaluated on the final 100 observations via a bootstrap predictive distribution (500 bootstrap draws). The objective (function to be minimized) is the median out-of-sample CRPS across all 1000 series. Hyperparameter search is carried out by Bayesian optimization (Moudiki, 2025a; Snoek et al., 2012) over the box – number of hidden nodes, number of lags, regularization parameter on direct link, regularization parameter on hidden layer – $[\![2, 10]\!] \times [\![1, 60]\!] \times [10^{-5}, 10^4]^2$ with 50 function evaluations. The resulting hyperparameters are then frozen (number of nodes in the hidden layer=7, number of lags=13, regularization parameter for the direct link=$10^{3.7}$, regularization parameter for the hidden layer=$10^{0.86}$) as well as the model's hidden layer weights (nodes), and embedded in a doubly-constrained ridge regression (see (Moudiki et al., 2018)) adjusted to training series lags. **No more retraining or cross-validation on real data required**; only ridge coefficients are estimated on each real series.

3. Results

Point forecasts are evaluated via median cross-validation Root Mean Squared Error (RMSE), and probabilistic forecasts via median CRPS, Coverage rate and Winkler score at a nominal level of 95 %. The median, here, guards against numerical failures in eGARCH estimation on volatile windows. Equivalence with eGARCH-t is assessed via two one-sided t-tests (TOST, Schuirmann, 1987). Hence, pretrained-rvfl2 is equivalent to eGARCH-t on RMSE ($p = 3.3e - 11$) and CRPS ($p = 9.5e - 06$) within margins of $\pm 5 \times 10^{-4}$ and $\pm 2 \times 10^{-4}$, representing approximately 4 % of each baseline value (Table 1). Equivalence is not established below 2 % of the baseline (see Table 5), confirming a small but detectable absolute difference. From 3 % onward it holds decisively for both metrics ($p < 10^{-4}$), placing the 4 % threshold used in Table 1 comfortably above the boundary. The non-conformal models over-cover by $\approx +2$ pp, consistent with the conservative tails of the student-t GARCH simulation, while **conformal wrappers generally bring coverage closer to the nominal 95 % level** (Table 2). pretrained-rvfl2 achieves a better mean Winkler95 skill score of +0.69 % (SD = 7.99 %) relative to eGARCH-t ($(W_{ref} - W95)/W_{ref}$, see Table 3). Notably, conformalization also tightens eGARCH-t intervals (Conf-eGARCH-t skill +0.06%) (Table 3). pretrained-rvfl2 requires only matrix multiplications on fixed pretrained weights and runs 10× faster than eGARCH-t (median wall time across 15 tickers). Conformal wrapping doubles execution time for both base learners: Conf-pretrained-rvfl2 is 3.6× faster than eGARCH-t, while Conf-eGARCH-t is 0.5× slower than eGARCH-t (eGARCH implementation from R package **rugarch**, see Table 4).

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Table 1: TOST equivalence tests vs eGARCH-t using a 4% baseline margin. Equiv. = Yes when TOST $p < 0.05$. See also Table 5.

Model	Metric	Mean diff.	90% CI	Margin	TOST p	Equiv.
pretrained-rvfl2	RMSE	1.6e-04	[9.0e-05, 2.2e-04]	5.0e-04	3.3e-11	Yes
pretrained-rvfl2	CRPS	8.8e-05	[4.9e-05, 1.3e-04]	2.0e-04	9.5e-06	Yes
Conf-eGARCH-t	RMSE	5.9e-07	[-3.5e-05, 3.6e-05]	5.0e-04	5.8e-27	Yes
Conf-eGARCH-t	CRPS	5.6e-05	[2.9e-05, 8.3e-05]	2.0e-04	7.9e-12	Yes

Table 2: Coverage averaged across tickers (in %) at a 95% nominal level by model and forecast horizon

Model	$h = 5$	$h = 10$	$h = 21$
eGARCH-t	100	97.00	96.19
pretrained-rvfl2	100	96.33	95.24
Conf-eGARCH-t	100	94.67	94.60
Conf-pretrained-rvfl2	100	94.00	93.97

Table 3: Winkler95 skill relative to eGARCH-t (positive = better Winkler score than eGARCH-t, tighter interval) and raw Winkler95 statistics.

Model	Mean skill (%)	SD skill (%)	Mean Winkler95	SD Winkler95
pretrained-rvfl2	0.6882	7.9867	0.0581	0.0137
Conf-eGARCH-t	0.4212	7.8596	0.0582	0.0144
eGARCH-t	0.0000	0.0000	0.0582	0.0116
Conf-pretrained-rvfl2	-0.2626	9.8095	0.0588	0.0161

Table 4: Median cross-validation computation time and speed-up factor vs eGARCH-t.

Model	Median time (s)	Speed-up
pretrained-rvfl2	6.3	9.6
Conf-pretrained-rvfl2	16.8	3.6
eGARCH-t	60.5	1.0
Conf-eGARCH-t	128.5	0.5

Table 5: Sensitivity of TOST equivalence (pretrained-rvfl2 vs eGARCH-t) to the choice of margin, expressed as a percentage of the eGARCH-t baseline mean. Equiv. = Yes when TOST $p < 0.05$.

Margin (% of baseline)	Metric	TOST p	Equiv.
1%	RMSE	8.4e-01	No
1%	CRPS	8.3e-01	No
2%	RMSE	3.3e-02	Yes
2%	CRPS	3.2e-02	Yes
3%	RMSE	9.4e-06	Yes
3%	CRPS	1.1e-05	Yes
4%	RMSE	5.3e-10	Yes
4%	CRPS	7.6e-10	Yes
5%	RMSE	5.1e-14	Yes
5%	CRPS	8.5e-14	Yes
6%	RMSE	1.4e-17	Yes
6%	CRPS	2.6e-17	Yes
7%	RMSE	1.1e-20	Yes
7%	CRPS	2.1e-20	Yes
8%	RMSE	1.9e-23	Yes
8%	CRPS	3.9e-23	Yes
9%	RMSE	7.0e-26	Yes
9%	CRPS	1.5e-25	Yes
10%	RMSE	4.6e-28	Yes
10%	CRPS	1.0e-27	Yes